

JACKSON C. TURNER

RESEARCH SCIENTIST | APPLIED MATHEMATICS, PHYSICS & SCIENTIFIC COMPUTING

New York, NY | me@jacktrnr.com | jacktrnr.com | github.com/jacktrnr

Experience

Harmoniqs

Brooklyn, NY

Quantum Research Scientist

Jun 2026 – Present

- Develop Julia simulation and optimization tools for neutral-atom transport using GitHub issues, PRs, and peer review
- Build an AI research assistant to help researchers navigate and apply technical knowledge
- Build fluency in QEC, neutral-atom hardware, and optical tweezers for quantum control
- Lead technical discussions with academic and industry partners; contribute to grants and investor materials

Atlas Fusion

Remote

Mathematical Physicist (Contract)

Feb – Jun 2026

- Built Julia numerical solvers for plasma-stability problems

Columbia University, APAM Department

New York, NY

Graduate Research Assistant

May 2022 – Jun 2026

- Established resonance-induced nonlinear bound states via scattering and bifurcation theory; *Nonlinearity*
- Built $< 10^{-8}$ -tolerance solvers for scattering, bifurcation, and dynamics; Simons speaker and 2025 Best Poster

Zap Energy, Theory & Modeling Division

Everett, WA

Research Intern

Jun – Aug 2025

- Built a Julia MHD-stability eigenvalue solver: $720\times$ faster (4 hours $\rightarrow$ 20 seconds); arclength continuation
- Identified a plasma-instability mechanism in sheared-flow Z-pinches (arXiv:2608.00291)

Education

Columbia University

June 2026

Ph.D. in Applied Mathematics

New York, NY

Engineering Presidential Fellow. Dissertation: *Resonance-induced nonlinear bound states and their orbital stability*. Advisor: M. I. Weinstein.

University of Utah

May 2021

Honors B.S. in Applied Mathematics & Physics

Salt Lake City, UT

Presidential Scholar.

Technical Skills

Programming Julia, Python (NumPy, SciPy), MATLAB

Software Development Git/GitHub, PR workflows, issue tracking, code review, reproducible research software

Scientific Computing Numerical PDEs, nonlinear eigenproblems, spectral methods, continuation, optimization

Physics Neutral-atom quantum computing and control, optical tweezers, magnetohydrodynamics, plasma stability

Mathematics Spectral and scattering theory, nonlinear waves, stability and bifurcation theory

Publications

- Crews, D. W. & J. C. Turner. “On the ideal stability of the sheared-flow Z pinch.” arXiv:2608.00291, 2026.
- Turner, J. C. & M. I. Weinstein. “Stability and instability analysis of resonance-induced nonlinear bound states.” arXiv:2610.01875, 2026.
- Turner, J. C. & M. I. Weinstein. “Resonance-induced nonlinear bound states.” *Nonlinearity* 39 (2026), 095030.
- Kao, C.-Y., B. Osting & J. C. Turner. “Flat Tori with Large Laplacian Eigenvalues.” *SIAM J. Appl. Algebra Geom.*, 2023.
- Turner, J. C., E. Cherkaev & D. Wang. “Expansion Method for Laplace–Beltrami Eigenfunctions.” arXiv:2210.10982, 2022.

Additional

Leadership & Languages Led teams of 10–20 volunteers across Hong Kong; fluent in Cantonese